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## The Comparison of Macronix Serial NOR Flash Products MX25L128xx6/F

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### 1. Introduction

This document compares the MX25L128356/MX25L128736 ("6" Version) and the MX25L12833F/MX25L12872F/MX25L12835F ("F" Version).

Generally, the "6" version and "F" version are pin and command compatible with the basic Read/Program/Erase commands. There may be some differences if special features are used such as Factory Mode, etc.

The document does not provide detailed information on the individual devices, but highlight the major similarities and differences between them. The comparison covers the general features, performance, command codes and other differences.

The information provided in this document is based on datasheets listed in Section ["9. Reference Documents"](#). Newer versions of the datasheets may override the contents of this document. A comparison of key features is provided in ["Table 1-1: Key Feature Comparison"](#).

Table 1-1: Key Feature Comparison

|                          |                                                                  |      | Part Number                 |                             |               |
|--------------------------|------------------------------------------------------------------|------|-----------------------------|-----------------------------|---------------|
| Feature                  |                                                                  |      | MX25L128356/<br>MX25L128736 | MX25L12833F/<br>MX25L12872F | MX25L12835F   |
| Technology               |                                                                  |      | 68nm                        | 75nm                        | 75nm          |
| Density                  |                                                                  |      | 128Mb                       | 128Mb                       | 128Mb         |
| VCC                      |                                                                  |      | 2.7V-3.6V                   | 2.7V-3.6V                   | 2.7V-3.6V     |
| Temperature              |                                                                  |      | -40°C to 85°C               | -40°C to 85°C               | -40°C to 85°C |
| Structure                |                                                                  |      |                             |                             |               |
| Operation<br>Frequency   | READ (1-1-1)                                                     |      | 50MHz                       | 50MHz                       | 50MHz         |
|                          | FAST READ (1-1-1)                                                |      | 133MHz                      | 133MHz                      | 133MHz        |
|                          | DREAD (1-1-2)                                                    |      | 133MHz                      | 133MHz                      | 133MHz        |
|                          | 2READ(1-2-2)                                                     |      | 133MHz                      | 133MHz                      | 133MHz        |
|                          | QREAD(1-1-4)                                                     |      | 133MHz                      | 133MHz                      | 133MHz        |
|                          | 4READ(1-4-4)                                                     |      | 133MHz@3.0V-3.6V            | 133MHz@3.0V-3.6V            | 133MHz        |
| Sector Size              |                                                                  |      | 4KB                         | 4KB                         | 4KB           |
| Block Size               |                                                                  |      | 32KB/64KB                   | 32KB/64KB                   | 32KB/64KB     |
| Program Buffer Size      |                                                                  |      | 256 bytes                   | 256 bytes                   | 256 bytes     |
| SFDP                     |                                                                  |      | Yes                         | Yes                         | Yes           |
| Endurance                |                                                                  | Typ. | 100K cycles                 | 100K cycles                 | 100K cycles   |
| Data Retention           |                                                                  | Typ. | 20 years                    | 20 years                    | 20 years      |
| tVSL                     | VCC (min) to CS# low<br>(VCC Rise Time)                          | Min. | 800us                       | 800us                       | 800us         |
| tDP                      | CS# High to Deep<br>Power-down Mode                              | Max. | 10us                        | 10us                        | 10us          |
| tRES1                    | CS# High to Standby<br>Mode without Electronic<br>Signature Read | Max. | 30us                        | 30us                        | 30us          |
| tRES2                    | CS# High to Standby<br>Mode with Electronic<br>Signature Read    | Max. | 30us                        | 30us                        | 30us          |
| Hardware Features        |                                                                  |      |                             |                             |               |
| RESET#                   |                                                                  |      | MX25L128356 only            | MX25L12833F only            | YES           |
| WP#                      |                                                                  |      | MX25L128356 only            | MX25L12833F only            | YES           |
| Software Features        |                                                                  |      |                             |                             |               |
| Suspend/Resume           |                                                                  |      | YES                         | YES                         | YES           |
| Deep Power Down Mode     |                                                                  |      | YES                         | YES                         | YES           |
| Secured OTP Size         |                                                                  |      | 4Kb x 2                     | 4Kb x 2                     | 4Kb           |
| Software Reset           |                                                                  |      | YES                         | YES                         | YES           |
| Quad Enable (QE)=1 fixed |                                                                  |      | MX25L128736 only            | MX25L12872F only            | -             |

## 2. Command Set

The core commands for read, erase, and program are major compatible between the "6" and "F" versions. For a full list of commands and a description of their functions, please refer to each product's datasheet.

**Table 2-1: Command Set Comparison**

| Command             |                 | Description                  | MX25L128356/<br>MX25L128736 | MX25L12833F/<br>MX25L12872F | MX25L12835F |
|---------------------|-----------------|------------------------------|-----------------------------|-----------------------------|-------------|
| Array<br>access     | READ            | Normal Read                  | 03                          | 03                          | 03          |
|                     | FAST READ       | Fast Read Data               | 0B                          | 0B                          | 0B          |
|                     | DREAD           | 1I 2O Read                   | 3B                          | 3B                          | 3B          |
|                     | 2READ           | 2I/O Read                    | BB                          | BB                          | BB          |
|                     | QREAD           | 1I 4O Read                   | 6B                          | 6B                          | 6B          |
|                     | 4READ           | 2I/O Read                    | EB                          | EB                          | EB          |
|                     | PP              | Page Program                 | 02                          | 02                          | 02          |
|                     | 4PP             | Quad Page Program            | 38                          | 38                          | 38          |
|                     | SE              | Sector Erase                 | 20                          | 20                          | 20          |
|                     | BE 32K          | Block Erase 32KB             | 52                          | 52                          | 52          |
|                     | BE              | Block Erase 64KB             | D8                          | D8                          | D8          |
|                     | CE              | Chip Erase                   | 60 or C7                    | 60 or C7                    | 60 or C7    |
| Device<br>operation | WREN            | Write Enable                 | 06                          | 06                          | 06          |
|                     | WRDI            | Write Disable                | 04                          | 04                          | 04          |
|                     | WPSEL           | Write Protect Selection      | 68                          | 68                          | 68          |
|                     | EQIO            | Enable QPI                   | 35                          | 35                          | 35          |
|                     | RSTQIO          | Reset QPI                    | F5                          | F5                          | F5          |
|                     | PGM/ERS Suspend | Suspends Program/ Erase      | 75 or B0                    | 75 or B0                    | B0          |
|                     | PGM/ERS Resume  | Resumes Program/ Erase       | 7A or 30                    | 7A or 30                    | 30          |
|                     | DP              | Deep Power Down              | B9                          | B9                          | B9          |
|                     | RDP             | Release from Deep Power Down | AB                          | AB                          | AB          |
|                     | NOP             | No Operation                 | 00                          | 00                          | 00          |
|                     | RSTEN           | Reset Enable                 | 66                          | 66                          | 66          |
|                     | RST             | Reset Memory                 | 99                          | 99                          | 99          |
|                     | GBLK            | Gang Block lock              | 7E                          | 7E                          | 7E          |
|                     | GBULK           | Gang Block unlock            | 98                          | 98                          | 98          |
|                     | FMEN            | Factory Mode Enable          | 41                          | 41                          | X           |

| Command            |         | Description                                 | MX25L128356/<br>MX25L128736 | MX25L12833F/<br>MX25L12872F | MX25L12835F |
|--------------------|---------|---------------------------------------------|-----------------------------|-----------------------------|-------------|
| Register<br>Access | RDID    | Read Identification                         | 9F                          | 9F                          | 9F          |
|                    | RES     | Read Electronic ID                          | AB                          | AB                          | AB          |
|                    | REMS    | Read Electronic Manufacturer &<br>Device ID | 90                          | 90                          | 90          |
|                    | QPIID   | QPI ID Read                                 | AF                          | AF                          | AF          |
|                    | RDSFDP  | Read SFDP Table                             | 5A                          | 5A                          | 5A          |
|                    | RDSR    | Read Status Register                        | 05                          | 05                          | 05          |
|                    | RDCR    | Read Configuration Register                 | 15                          | 15                          | 15          |
|                    | RDFMSR  | Read Factory Mode Status Register           | 44                          | X                           | X           |
|                    | WRSR    | Write Status/Configuration Register         | 01                          | 01                          | 01          |
|                    | RDSCUR  | Read Security Register                      | 2B                          | 2B                          | 2B          |
|                    | WRSCUR  | Write Security Register                     | 2F                          | 2F                          | 2F          |
|                    | SBL     | Set urst Length                             | C0                          | C0                          | C0          |
|                    | ENSO    | Enter Secured OTP                           | B1                          | B1                          | B1          |
|                    | EXSO    | Exit Secured OTP                            | C1                          | C1                          | C1          |
|                    | WRLR    | Write Lock Register                         | 2C                          | 2C                          | 2C          |
|                    | RDLR    | Read Lock Register                          | 2D                          | 2D                          | 2D          |
|                    | WRSPB   | SPB Bit Program                             | E3                          | E3                          | E3          |
|                    | ESSPB   | All SPB Bit Erase                           | E4                          | E4                          | E4          |
|                    | RDSPB   | Read SPB Status                             | E2                          | E2                          | E2          |
|                    | WRDPB   | Write DPB Register                          | E1                          | E1                          | E1          |
|                    | RDDPB   | Read DPB Register                           | E0                          | E0                          | E0          |
|                    | RDFBR   | Read Fast Boot Register                     | X                           | X                           | 16          |
|                    | WRFBR   | Write Fast Boot Register                    | X                           | X                           | 17          |
|                    | ESFBR   | Erase Fast Boot Register                    | X                           | X                           | 18          |
|                    | WRPASS  | Write Password Register                     | X                           | X                           | 28          |
|                    | RDPASS  | Read Password Register                      | X                           | X                           | 27          |
|                    | PASSULK | Password Unlock                             | X                           | X                           | 29          |
|                    | SPBLK   | SPB Lock Set                                | X                           | X                           | A6          |
|                    | RDSPBLK | SPB Lock Register Read                      | X                           | X                           | A7          |

### 3. Data Protection

Both "6" and "F" version products provide the same protection modes to protect sectors.

#### 3-1. BP bit Block Protection

All of the listed Macronix flash can use BP bits in the Status Register to select groups of memory areas for write protection. All of the regions protected by "F" version can be protected by "6" version with the identical BP setting. Please refer to the following comparison table.

**Table 3-1: Block Protection (BP) Comparison of MX25L128xx6/F**

**Protected Area Sizes (T/B bit = 0)**

| Status bit |     |     |     | Protect Level                                                   |
|------------|-----|-----|-----|-----------------------------------------------------------------|
| BP3        | BP2 | BP1 | BP0 | MX25L128356/MX25L128736/<br>MX25L12833F/MX25L12872F/MX25L12835F |
| 0          | 0   | 0   | 0   | None                                                            |
| 0          | 0   | 0   | 1   | 1block (#255)                                                   |
| 0          | 0   | 1   | 0   | 2blocks (#254-255)                                              |
| 0          | 0   | 1   | 1   | 4blocks (#252-255)                                              |
| 0          | 1   | 0   | 0   | 8blocks (#248-255)                                              |
| 0          | 1   | 0   | 1   | 16blocks (#240-255)                                             |
| 0          | 1   | 1   | 0   | 32blocks (#224-255)                                             |
| 0          | 1   | 1   | 1   | 64blocks (#192-255)                                             |
| 1          | 0   | 0   | 0   | 128blocks (#128-255)                                            |
| 1          | 0   | 0   | 1   | 256blocks (all)                                                 |
| 1          | 0   | 1   | 0   | 256blocks (all)                                                 |
| 1          | 0   | 1   | 1   | 256blocks (all)                                                 |
| 1          | 1   | 0   | 0   | 256blocks (all)                                                 |
| 1          | 1   | 0   | 1   | 256blocks (all)                                                 |
| 1          | 1   | 1   | 0   | 256blocks (all)                                                 |
| 1          | 1   | 1   | 1   | 256blocks (all)                                                 |

## 4. Register Comparison

The MX25L128356/MX25L128736 and MX25L12833F/MX25L12872F/MX25L12835F Status Register bits are compatible.

**Table 4-1: Status Register Bit Comparison**

|      | <b>MX25L128356/MX25L12833F/MX25L12835F</b> | <b>MX25L128736/MX25L12872F</b> |
|------|--------------------------------------------|--------------------------------|
| Bit0 | Write in Progress Bit                      | Write in Progress Bit          |
| Bit1 | Write Enable Latch                         | Write Enable Latch             |
| Bit2 | Block Protected: BP0                       | Block Protected: BP0           |
| Bit3 | Block Protected: BP1                       | Block Protected: BP1           |
| Bit4 | Block Protected: BP2                       | Block Protected: BP2           |
| Bit5 | Block Protected: BP3                       | Block Protected: BP3           |
| Bit6 | Quad Enable                                | Quad Enable                    |
| Bit7 | Status Register Write Protect              | Reserved                       |

**Table 4-2: Configuration Register Bit Comparison**

|      | <b>MX25L128356/MX25L128736/<br/>MX25L12833F/MX25L12872F/MX25L12835F</b> |
|------|-------------------------------------------------------------------------|
| Bit0 | Output Driver Strength: ODS0                                            |
| Bit1 | Output Driver Strength: ODS1                                            |
| Bit2 | Output Driver Strength: ODS2                                            |
| Bit3 | Top/Bottom selected                                                     |
| Bit4 | Reserved                                                                |
| Bit5 | Reserved                                                                |
| Bit6 | Dummy Cycles 0                                                          |
| Bit7 | Dummy Cycles 1                                                          |

**Table 4-3: Security Register Comparison**

|      | <b>MX25L128356/MX25L128736/<br/>MX25L12833F/MX25L12872F/MX25L12835F</b> |
|------|-------------------------------------------------------------------------|
| Bit0 | Secured OTP Indicator Bit                                               |
| Bit1 | Lock-down Secured OTP                                                   |
| Bit2 | Program Suspend Status                                                  |
| Bit3 | Erase Suspend Status                                                    |
| Bit4 | Reserved                                                                |
| Bit5 | P_FAIL                                                                  |
| Bit6 | E_FAIL                                                                  |
| Bit7 | WPSEL                                                                   |

## 5. Electrical Characteristics

The comparison of DC and AC characteristics are shown in Tables 5-1 and 5-2:

**Table 5-1: DC Performance Comparison**

| DC Performance                           |      | MX25L128356/<br>MX25L128736 | MX25L12833F/<br>MX25L12872F | MX25L12835F |
|------------------------------------------|------|-----------------------------|-----------------------------|-------------|
| VCC Standby Current (ISB1)               | Typ. | 10uA                        | 10uA                        | 10uA        |
|                                          | Max. | 50uA                        | 50uA                        | 50uA        |
| Deep Power Down Current (ISB2)           | Typ. | 2uA                         | 2uA                         | 2uA         |
|                                          | Max. | 20uA                        | 20uA                        | 20uA        |
| VCC Read Current (ICC1) (4IO)            | Typ. | 14mA                        | 14mA                        | 14mA        |
|                                          | Max. | 25mA                        | 25mA                        | 25mA        |
| VCC Program Current (ICC2)               | Typ. | 14mA                        | 14mA                        | 14mA        |
|                                          | Max. | 20mA                        | 20mA                        | 20mA        |
| VCC Write Status Register Current (ICC3) | Typ. | 10mA                        | 10mA                        | 10mA        |
|                                          | Max. | 12mA                        | 12mA                        | 12mA        |
| VCC Sector Erase Current (ICC4)          | Typ. | 14mA                        | 14mA                        | 14mA        |
|                                          | Max. | 25mA                        | 25mA                        | 25mA        |
| VCC Chip Erase Current (ICC5)            | Typ. | 14mA                        | 14mA                        | 14mA        |
|                                          | Max. | 25mA                        | 25mA                        | 25mA        |

**Table 5-2: AC Performance Comparison**

| AC Performance                    |            |      | MX25L128356/<br>MX25L128736     | MX25L12833F/<br>MX25L12872F | MX25L12835F |
|-----------------------------------|------------|------|---------------------------------|-----------------------------|-------------|
| Byte Program Time                 |            | Typ. | 10us                            | 16us                        | 16us        |
|                                   |            | Max. | 280us                           | 40us                        | 30us        |
| Page Program Time                 | 256 Bytes  | Typ. | 0.33ms                          | 0.33ms                      | 0.5ms       |
|                                   |            | Max. | 2.4ms                           | 1.2ms                       | 1.5ms       |
| Erase Time                        | 4KB        | Typ. | 25ms                            | 25ms                        | 30ms        |
|                                   |            | Max. | 400ms                           | 120ms                       | 120ms       |
|                                   | 32KB       | Typ. | 0.14s / 0.05s <sup>(Note)</sup> | 0.14s                       | 0.15s       |
|                                   |            | Max. | 0.85s                           | 0.65s                       | 0.65s       |
|                                   | 64KB       | Typ. | 0.25s / 0.09s <sup>(Note)</sup> | 0.25s                       | 0.28s       |
|                                   |            | Max. | 1.6s                            | 0.65s                       | 0.65s       |
|                                   | Chip Erase | Typ. | 12s <sup>(Note)</sup>           | 26s                         | 50s         |
|                                   |            | Max. | 60s                             | 60s                         | 80s         |
| Write Status Register Cycle Time  |            | Max. | 40ms                            | 40ms                        | 40ms        |
| tCLQV (Clock Low to Output Valid) | 15pF       | Max. | 6ns                             | 6ns                         | 6ns         |
|                                   | 30pF       | Max. | 8ns                             | 8ns                         | 8ns         |

**Note:** Blank to Blank pattern.

## 6. Memory Organization

The memory and sector architecture of the MX25L128xx6/F flash memory are identical.

## 7. Device Identification

The Manufacturer ID and Device ID values of the G version flash memory are identical to the F version flash memory.

**Table 7-1: ID Code Comparison**

| Electronic Identification |     |                 | MX25L128356/MX25L128736/<br>MX25L12833F/MX25L12872F/MX25L12835F |
|---------------------------|-----|-----------------|-----------------------------------------------------------------|
| RDID                      | 9Fh | Manufacturer ID | C2                                                              |
|                           |     | Type            | 20                                                              |
|                           |     | Density         | 18                                                              |
| RES                       | ABh | Electronic ID   | 17                                                              |
| REMS                      | 90h | Manufacturer ID | C2                                                              |
|                           |     | Device ID       | 17                                                              |

## 8. Summary

This document compares MX25L128356/MX25L128736 ("6" Version) and MX25L12833F/MX25L12872F/MX25L12835F ("F" Version).

Generally, the "6" version and "F" version are pin and command compatible with the basic Read/Program/Erase commands. There may be some differences if special features are used such as Factory Mode, etc.

The document does not provide detailed information on the individual devices, but highlight the major similarities and differences between them. For more details, please check the original datasheets, the content of which may override this application note if there is any inconsistency.

## 9. Reference Documents

Table 9-1 shows the datasheet versions used for comparison in this application note. For the most current Macronix specification, please refer to the Macronix Website at <http://www.macronix.com>

**Table 9-1: Datasheet Version**

| Datasheet   | Location         | Date Issued        | Versions  |
|-------------|------------------|--------------------|-----------|
| MX25L128356 | Macronix Website | August 09, 2022    | Rev. 1.0  |
| MX25L128736 | Macronix Website | September 02, 2022 | Rev. 0.01 |
| MX25L12833F | Macronix Website | October 17, 2017   | Rev 1.0   |
| MX25L12872F | Macronix Website | April 09, 2021     | Rev. 1.1  |
| MX25L12835F | Macronix Website | July 22, 2016      | Rev 1.6   |

## 10.Revision History

**Table 10-1: Revision History**

| Revision No. | Description                                                                                                                                                                     | Page     | Date               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------|
| Rev. 1       | Initial Release                                                                                                                                                                 | ALL      | April 27, 2022     |
| Rev. 2       | 1. Description modification<br>2. Modified the tBP, tPP, tSE, tBE32K, tBE, tCE values of MX25L128356 and MX25L128736 and added notes<br>3. Updated Table 9-1: Datasheet Version | P2, 7, 9 | September 14, 2022 |



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